Before we move on to look at use cases for Elasticsearch, let's briefly go over these essential moving parts at a high level. Other than Elasticsearch, we do not discuss this stack of products in this book.

BEATS

Beats are single purpose data shippers; they load data from out various external systems and pump it into Elasticsearch. Various types of beats are available out of the box. These include Filebeat, Metricbeat, Heartbeat, etc., and each performs a specific data consumption task. These are single-purpose components: for example, Filebeats are designed for file-based transports and Metricbeats for vital machine and operating system memory and CPU information. The beats' agents are installed on the servers so they can consume data from their source and send it to their destination.

LOGSTASH

Logstash is an open source data-processing engine. It extracts data originating from multiple sources, processes it, and sends it to a variety of target destinations. During the processing of the data, Logstash transforms and enriches the data. It supports a myriad of sources and destinations including files, HTTP, JMS, Kafka, Amazon S3, Twitter, and dozens of others. It promotes a pipeline architecture, and every event that goes through the pipeline is parsed as per the preconfigured rules, thus creating a real-time pipeline for data ingestion.

KIBANA

Kibana is a multipurpose web console that provides a host of options such as executing queries; developing dashboards, graphs, and chart visualizations; and creating dropdowns and aggregations. However, we can use any REST client to talk to Elasticsearch to invoke the APIs, not just Kibana. For example, we can invoke APIs using cURL, Postman, or native language clients.

1.4.3 Elasticsearch use cases

Pinpointing Elasticsearch for a particular use case or domain is difficult. It is omnipresent in many areas from search to analytics to machine learning jobs. It is widely used across a multitude of industries, including finance, defense, transport, government, retail, cloud, entertainment, space, and more. Let's take a high-level glance at how Elasticsearch can be used in an organization.

SEARCH ENGINE

Elasticsearch has become the go-to technology for its full-text search capabilities. The product is not limited to full-text searching but can also be used for structured data and geolocation-based searches. Broadly speaking, customers use Elasticsearch in three domains: App Search, Enterprise Search, and Site Search.

In *App Search,* Elasticsearch serves as a backbone, providing search and analytical capabilities for applications. A search service backed up by Elasticsearch can be designed as a microservice that serves the application's search requirements, such as searching for customers, orders, invoices, emails, and so on.